P1.5 Kinetic energy and elastic energy stores

Learning objectives

After this topic, you should know:

- what the amount of energy in a kinetic energy store depends on
- how to calculate the amount of energy in a kinetic energy store
- what an elastic potential energy store is
- how to calculate the amount of energy in an elastic potential energy store.

The energy an object has because of its motion depends on its mass and speed. This energy is called kinetic energy.

Investigating kinetic energy stores

Figure 1 shows how you can investigate how the kinetic energy store of an object depends on its speed.

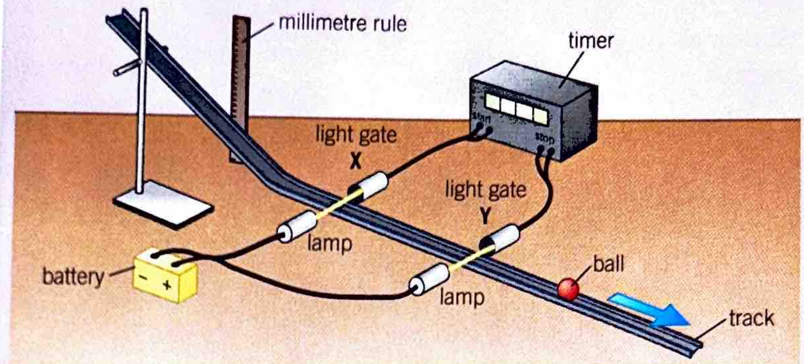

Figure 1 *Investigating changes in kinetic energy stores*

1 The ball is released on a slope from a measured height above the foot of the slope. You can calculate the decrease in its gravitational potential energy store by using the following equation:

change in gravitational potential energy store = mass × gravitational field strength × change in height.

Due to conservation of energy, this decrease in the gravitational potential energy store is matched by an equal increase in its kinetic energy store.

2 The ball is timed, using light gates, over a measured distance between X and Y after the slope.

- Explain why light gates improve the quality of the data you can collect in this investigation.

Investigating a catapult

Use rubber bands to catapult a trolley along a horizontal runway. Find out how the speed of the trolley depends on how much the catapult is pulled back before the trolley is released. For example, see if the distance needs to be doubled to double the speed. Figure 1 shows how the speed of the trolley can be measured.

Safety: Take care to ensure you do this safely. Protect your hands and feet, and the bench, from falling trolleys.

Table 1 shows some sample results.

Table 1 *Sample measurements for a ball of mass 0.5 kg*

Height drop to foot of slope in m	0.05	0.10	0.16	0.20
Initial kinetic energy of ball in J	0.25	0.50	0.80	1.00
Time to travel 1.0 m from X to Y in s	0.98	0.72	0.57	0.50
Average speed of ball between X and Y in m/s	1.02			2.00

Work out the speed of the ball between X and Y in each case. The first and last values have been worked out for you. Can you see a link between speed and height drop? The results show that the greater the height drop, the faster is the speed. So it can be said that the kinetic energy store of the ball increases if the speed increases.

The kinetic energy equation

Table 1 shows that when the height drop is increased by four times from 0.05 m to 0.20 m, the speed doubles. The height drop is directly proportional to the speed squared, or (speed)^2. Because the height drop is a measure of the ball's kinetic energy store, it can be said that the ball's kinetic energy store is directly proportional to the square of its speed.

The amount of energy in the kinetic energy store of an object can be calculated using the kinetic energy equation below:

kinetic energy, E_k = $\frac{1}{2}$ × **mass, m** × **speed^2, v^2**

(joules, J) (kilograms, kg) (metres per second, m/s)^2

Using elastic potential energy

When you stretch a rubber band or a spring, the work you do is stored in it as **elastic potential energy**.

Figure 2 shows how the force F needed to stretch a spring varies with its extension e. The graph obeys the equation for Hooke's Law $F = k\ e$, where k is the spring constant.

For a spring stretched to an extension e, we can calculate the energy in its elastic potential energy store using the equation below:

elastic potential energy, E_e = $\frac{1}{2}$ × **spring constant, k** × **extension^2, e^2**

(joules, J) (newtons per metre, N/m) (metres, m)^2

1 a Calculate the kinetic energy store of:
i a vehicle of mass 500 kg moving at a speed of 12 m/s [2 marks]
ii a football of mass 0.44 kg moving at a speed of 20 m/s. [2 marks]
b Calculate the velocity of a 500 kg vehicle with twice the kinetic energy store as calculated in **a i**. [3 marks]

2 a A catapult is used to fire an object into the air. Describe the energy transfers when the catapult is:
i stretched [2 marks] **ii** released. [2 marks]
b An object of weight 2.0 N fired vertically upwards from a catapult reaches a maximum height of 5.0 m. Calculate:
i the increase in the gravitational potential energy store of the object [1 mark]
ii the speed of the object when it left the catapult. [4 marks]

3 A car moving at a constant speed has 360 000 J in its kinetic energy store. When the driver applies the brakes, the car stops in a distance of 100 m.
a Calculate the force that stops the vehicle. [3 marks]
b The speed of the car was 30 m/s when its kinetic energy store was 360 000 J. Calculate its mass. [3 marks]

4 A mobility aid to assist walking uses a steel spring to store energy when the walker's foot goes down, and it returns energy as the foot is lifted. The spring has a spring constant of 250 N/m. Calculate the elastic potential energy stored in the spring when its extension is 0.21 m. [2 marks]

Worked example

Calculate the kinetic energy stored in a vehicle of mass 500 kg moving at a speed of 12 m/s.

Solution

Kinetic energy $= \frac{1}{2} m v^2$
$= 0.5 \times 500\,\text{kg} \times (12\,\text{m/s})^2$
$= \mathbf{36\,000\,J}$

Go further!

In Figure 2, the force F increases as the extension e is increased. The average force when the spring is extended to extension e is $\frac{1}{2}F$, where $F = ke$. Therefore, the energy stored in the spring = work done = average force extension $= \frac{1}{2}F\,e = \frac{1}{2}ke^2$.

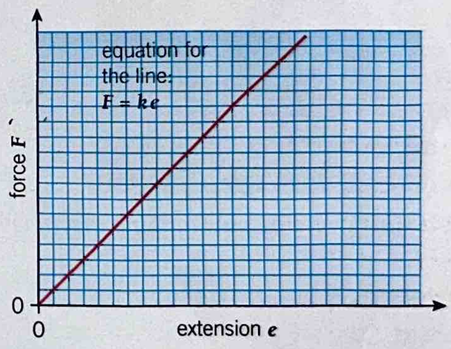

Figure 2 *Force versus extension for a spring. The spring constant k is the force per unit extension of the spring*

Key points

- The energy in the kinetic energy store of a moving object depends on its mass and its speed.
- The kinetic energy store of an object is $E_k = \frac{1}{2}mv^2$
- Elastic potential energy is the energy stored in an elastic object when work is done on the object.
- The elastic potential energy stored in a stretched spring is $E_e = \frac{1}{2}ke^2$, where e is the extension of the spring.